

CSE 3203 – Software Engineering

Lecture 03: Introduction to SE (Part 2)

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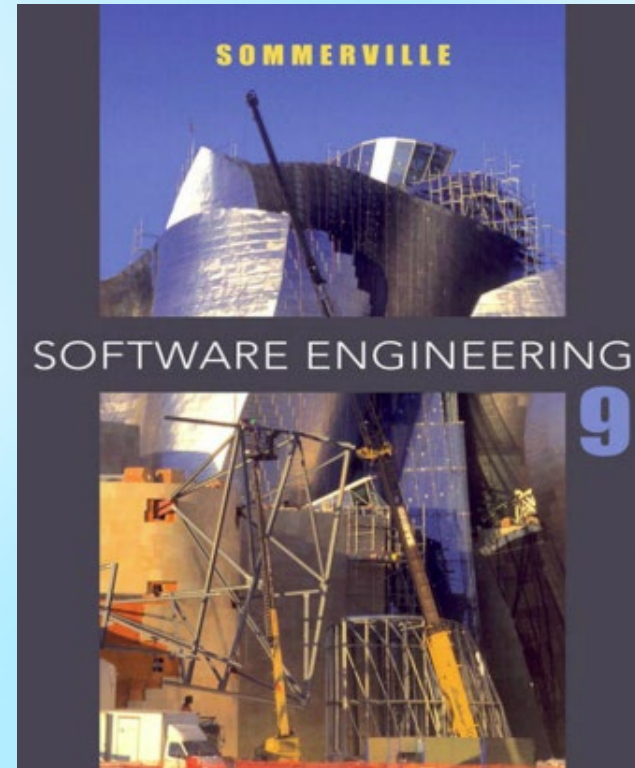
Class Code
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Book for Reference

- Software Engineering 9th Edition by Ian Sommerville



Mark Distribution



- Mid Semester Exam – 25%
- Final Semester Exam – 40%
- Class Attendance & Performance – 10%
- Course Project – 25%

Is it possible to get A/A+ ?

- Absolutely. Just follow my instructions to the point.

How to fail in this course ?

- Don't follow my instructions.

Instructions



Software engineering diversity

- There are many different types of software system and there is no universal set of software techniques that is applicable to all of these.
- The most significant factor: the type of application.

Application types

- Stand-alone applications
 - Run on a local computer, such as a PC, include all necessary functionality and do not need to be connected to a network.
- Interactive transaction-based applications
 - Execute on a remote computer and accessed by users from their own PCs or terminals.
- Embedded control systems
 - Control and manage hardware devices.

Application types

- Batch processing systems
 - Designed to process data in large batches. They process large numbers of individual inputs to create corresponding outputs.
- Entertainment systems
 - Personal use and intended to entertain the user.
- Systems for modelling and simulation
 - Developed by scientists and engineers to model physical processes or situations, which include many, separate, interacting objects.

Application types

- Data collection systems
 - Collect data from their environment using a set of sensors and send that data to other systems for processing.
- Systems of systems
 - These are systems that are composed of a number of other software systems.

Software engineering fundamentals

- Some fundamental principles apply to all types of software system:
- Systems should be developed using a managed and understood development process.
- Dependability and performance are important for all types of system.
- Understanding and managing the software specification and requirements.
- Where appropriate, you should reuse software that has already been developed rather than write new software.

Software engineering and the web

- The Web is now a platform for running application and organizations are increasingly developing web-based systems rather than local systems.
- Web services allow application functionality to be accessed over the web.
- Cloud computing is an approach to the provision of computer services where applications run remotely on the 'cloud'.
 - Users do not buy software but pay according to use.

Web software engineering

- Software reuse is the dominant approach for constructing web-based systems.
- Web-based systems should be developed and delivered incrementally.
- User interfaces are constrained by the capabilities of web browsers.
 - Technologies such as AJAX allow rich interfaces to be created within a web browser but are still difficult to use. Web forms with local scripting are more commonly used.

Web-based software engineering

- Web-based systems are complex distributed systems but the fundamental principles of software engineering discussed previously are as applicable to them as they are to any other types of system.
- The fundamental ideas of software engineering, discussed in the previous section, apply to web-based software in the same way that they apply to other types of software system.

Key points

- Software engineering is an engineering discipline that is concerned with all aspects of software production.
- Essential software product attributes are maintainability, dependability and security, efficiency and acceptability.
- The high-level activities of specification, development, validation and evolution are part of all software processes.
- The fundamental notions of software engineering are universally applicable to all types of system development.

Key points

- There are many different types of system and each requires appropriate software engineering tools and techniques for their development.
- The fundamental ideas of software engineering are applicable to all types of software system.



Thank You

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